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10 CFR 50.73

July 21, 2017
NRC-17-0048

U. S. Nuclear Regulatory Commission
Attention: Document Control Desk
Washington, D.C. 20555-0001

Reference: Fermi 2
NRC Docket No. 50-341
NRC License No. NPF-43

Subject: Licensee Event Report (LER) No. 2017-003

Pursuant to 10 CFR 50.73(a)(2)(i)(B) and 10 CFR 50.73(a)(2)(v)(B), DTE Electric Company (DTE) is submitting LER No. 2017-003, Division 2 Residual Heat Removal Service Water System (RHRSW) Inoperable Due to an Inoperable RHRSW Flow Control Valve.

No new commitments are being made in this LER.

Should you have any questions or require additional information, please contact Mr. Scott A. Maglio, Manager Nuclear Licensing, at (734) 586-5076.

Sincerely,

Keith J. Polson
Site Vice President

Enclosure: Licensee Event Report No. 2017-003, Division 2 Residual Heat Removal Service Water System (RHRSW) Inoperable Due to an Inoperable RHRSW Flow Control Valve

cc: NRC Project Manager
NRC Resident Office
Reactor Projects Chief, Branch 5, Region III
Regional Administrator, Region III
Michigan Public Service Commission
Regulated Energy Division (kindschl@michigan.gov)

**Enclosure to
NRC-17-0048**

**Fermi 2 NRC Docket No. 50-341
Operating License No. NPF-43**

**Licensee Event Report No. 2017-003, Division 2 Residual Heat Removal Service Water
System (RHRSW) Inoperable Due to an Inoperable RHRSW Flow Control Valve**

**LICENSEE EVENT REPORT (LER)**

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form)

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Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Privacy and Information Collections Branch (T-5 F53), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollections.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOB-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

1. FACILITY NAME

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2. DOCKET NUMBER

05000 341

3. PAGE

1 OF 6

4. TITLE Division 2 Residual Heat Removal Service Water System (RHRSW) Inoperable Due to an Inoperable RHRSW Flow Control Valve

5. EVENT DATE			6. LER NUMBER			7. REPORT DATE			8. OTHER FACILITIES INVOLVED	
MONTH	DAY	YEAR	YEAR	SEQUENTIAL NUMBER	REV NO.	MONTH	DAY	YEAR	FACILITY NAME	DOCKET NUMBER
05	22	2017		2017-003-00		07	21	2017	N/A	05000
									N/A	05000
9. OPERATING MODE			11. THIS REPORT IS SUBMITTED PURSUANT TO THE REQUIREMENTS OF 10 CFR §: (Check all that apply)							
1			☐ 20.2201(b)		☐ 20.2203(a)(3)(i)		☐ 50.73(a)(2)(ii)(A)		☐ 50.73(a)(2)(viii)(A)	
			☐ 20.2201(d)		☐ 20.2203(a)(3)(ii)		☐ 50.73(a)(2)(ii)(B)		☐ 50.73(a)(2)(viii)(B)	
			☐ 20.2203(a)(1)		☐ 20.2203(a)(4)		☐ 50.73(a)(2)(iii)		☐ 50.73(a)(2)(ix)(A)	
			☐ 20.2203(a)(2)(i)		☐ 50.36(c)(1)(i)(A)		☐ 50.73(a)(2)(iv)(A)		☐ 50.73(a)(2)(x)	
10. POWER LEVEL 100			☐ 20.2203(a)(2)(ii)		☐ 50.36(c)(1)(ii)(A)		☐ 50.73(a)(2)(v)(A)		☐ 73.71(a)(4)	
			☐ 20.2203(a)(2)(iii)		☐ 50.36(c)(2)		☒ 50.73(a)(2)(v)(B)		☐ 73.71(a)(5)	
			☐ 20.2203(a)(2)(iv)		☐ 50.46(a)(3)(ii)		☐ 50.73(a)(2)(v)(C)		☐ 73.77(a)(1)	
			☐ 20.2203(a)(2)(v)		☐ 50.73(a)(2)(i)(A)		☐ 50.73(a)(2)(v)(D)		☐ 73.77(a)(2)(i)	
			☐ 20.2203(a)(2)(vi)		☒ 50.73(a)(2)(i)(B)		☐ 50.73(a)(2)(vii)		☐ 73.77(a)(2)(ii)	
					☐ 50.73(a)(2)(i)(C)		☐ OTHER Specify in Abstract below or in NRC Form 366A			

12. LICENSEE CONTACT FOR THIS LER

LICENSEE CONTACT

Fermi 2 / Scott A. Maglio – Manager, Nuclear Licensing

TELEPHONE NUMBER (Include Area Code)
(734) 586-5076**13. COMPLETE ONE LINE FOR EACH COMPONENT FAILURE DESCRIBED IN THIS REPORT**

CAUSE	SYSTEM	COMPONENT	MANUFACTURER	REPORTABLE TO EPIX	CAUSE	SYSTEM	COMPONENT	MANUFACTURER	REPORTABLE TO EPIX
D	BI	FCV	P305	N	N/A	N/A	N/A	N/A	N/A

14. SUPPLEMENTAL REPORT EXPECTED☐ YES (If yes, complete 15. EXPECTED SUBMISSION DATE) ☒ NO**15. EXPECTED SUBMISSION DATE**

MONTH	DAY	YEAR

ABSTRACT (Limit to 1400 spaces, i.e., approximately 15 single-spaced typewritten lines)

On May 22, 2017 at 05:10 am (EST), while placing Division 2 Residual Heat Removal Service Water (RHRSW) in service for biocide treatment of the Division 2 Residual Heat Removal (RHR) Reservoir, the Division 2 RHRSW Flow Control Valve (FCV) (E1150F068B) failed to fully open. Troubleshooting discovered the direct cause was failure of the anti-rotation bushing stem key. The apparent cause was system operating conditions (high vibration) resulting in the failed tack welds. Previous troubleshooting on an indication issue on May 5, 2017 for the RHRSW FCV was inadequate, and did not identify the failure of the anti-rotation key. As a result, the RHRSW FCV was returned to service at 2:50 pm on May 7, 2017, and subsequently failed on the next on-demand stroke at 5:10 am on May 22, 2017. Seventeen similar Motor Operated Valves (MOVs) were inspected and no MOVs exhibiting the symptoms observed on the E1150F068B prior to the failure of the anti-rotation key were found, and all anti-rotation devices were found to be intact. The Past Operability determination for E1150F068B found that the MOV was unable to perform its design basis functions from May 3, 2017 at 5:48 am, when the RHRSW FCV was last successfully stroked under dynamic conditions, through May 24, 2017 at 4:04 pm, when the RHRSW FCV was returned to service. The Division 1 RHRSW was available throughout the event except on two occasions. Division 1 of RHRSW was declared inoperable for Mechanical Draft Cooling Tower (MDCT) Nozzle Cleaning activities on May 9, 2017 from 8:41 am to May 9, 2017 at 11:18 pm. Division 1 of RHRSW was again declared inoperable for MDCT Nozzle Cleaning activities on May 11, 2017 at 8:35 am through May 11, 2017 at 10:01 pm. The as found condition of the Division 2 RHRSW FCV is a condition prohibited by Technical Specification 3.7.1 and reportable under 10 CFR 50.73 (a)(2)(i)(B) "Operation or Condition Prohibited by Technical Specifications," and 10 CFR 50.73(a)(2)(v)(B) "Any event or condition that could have prevented the fulfillment of the safety function of structures or systems that are needed to: Remove Residual Heat."

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

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		YEAR 2017	SEQUENTIAL NUMBER -003	REV NO. -00

NARRATIVE**INITIAL PLANT CONDITIONS**

Mode – 1

Reactor Power – 100

Reactor Pressure Vessel Temperature – 518 degrees Fahrenheit

There were no structures, systems, or components (SSCs) that were inoperable at the start of this event that contributed to this event.

DESCRIPTION OF THE EVENT

On May 22, 2017, at 5:10 am, while placing Division 2 Residual Heat Removal Service Water (RHRSW) [BI] in service to support biocide treatment of the Division 2 Residual Heat Removal (RHR) [BO] Reservoir, the Division 2 RHR Heat Exchanger Service Water Flow Control Valve [(FCV)] (E1150F068B) did not completely open as expected when the push button was pressed. The Division 2 RHRSW Flow Control Valve (FCV) (E1150F068B) failed to fully open. The RHRSW FCV indicated dual position after 5 seconds of operation in the open direction. When the RHRSW pump was started, indicated flow would not go above 2800 Gallons Per Minute (gpm). After the initial flow spike RHRSW flow settled out at 2,800 gpm and would not continue to rise as expected while holding the open pushbutton for E1150F068B valve. The Division 2 RHRSW Pump B was placed in off and valve position remained in dual indication. The Operator pushed the close pushbutton and no change in valve position was observed. The operator then depressed the Thermal Overload reset and verified continuity for all fuses in 72F-4A Position 3A, MCC for E1150F068B. The FCV was verified closed locally by stem position and manually checking close with hand-wheel. Troubleshooting discovered the direct cause was failure of the anti-rotation bushing stem key tack welds. The welds were repaired and the key was replaced. Post Maintenance Testing (PMT) was satisfactorily completed. The RHRSW FCV was returned to service at 4:04 pm on May 24, 2017.

Prior to the May 22, 2017, RHRSW FCV failure, troubleshooting did not identify cause of the failure.

On May 5, 2017, during shutdown of the Division 2 RHRSW, with flow lowered to 8500 gpm, the E1150F068B improperly indicated full closed. The RHRSW pumps were shutdown and E1150F068B was then stroked and timed with local position verification. When the open pushbutton was pressed, the open light did not come on until after approximately 55 seconds and the valve indicated full open after 125 seconds. On the close stroke, the open light went out after approximately 56 seconds (indicated full closed in the Main Control Room) and the valve was seen to be full closed locally at 124 seconds. The stroke times were longer than expected but still within acceptance criteria.

Troubleshooting activities were performed. The troubleshooting focused on the indication aspects of the Motor

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Operated Valve (MOV), which involved inspecting the Torque Switch, Limit Switch, and the associated wiring. Hand wheel operation of the MOV was performed, and it was noted that the Limit Switches were not actuating when expected. The 4 train-limit switch assembly was replaced, and limits were reset.

On May 7, 2017 PMT was performed which included Equivalent Thrust diagnostic testing and Inservice Testing (IST) Stroke Time Testing Partial Surveillance 24.205.06. This PMT was performed at static conditions and in accordance with MMA11 "Post Maintenance Testing Guidelines." Due to the static versus dynamic nature of the testing, the actual failure mechanism was missed and the results were accepted as satisfactory with the valve being returned to service.

The RHRSW FCV E1150F068B and Division 2 RHRSW Subsystem were considered inoperable from May 3, when the RHRSW FCV was last successfully stroked under dynamic conditions, to May 24, 2017. This is a condition prohibited by Technical Specification (TS) 3.7.1 and reportable under 10 CFR 50.73 (a)(2)(i)(B) "Operation or Condition Prohibited by Technical Specifications." Also, during this time, Division 1 RHRSW subsystem was inoperable on two occasions. This is additionally reportable under 10 CFR 50.73(a)(2)(v)(B) "Any event or condition that could have prevented the fulfillment of the safety function of structures or systems that are needed to: Remove Residual Heat."

SIGNIFICANT SAFETY CONSEQUENCES AND IMPLICATIONS

There were no safety consequences or radiological releases associated with this event. At no time during this event was there a potential for endangering the public health and safety.

UFSAR 3.8.4.1.2

The RHR complex is a reinforced-concrete structure designed to serve as the ultimate heat sink for the reactor during normal shutdowns and postulated accident conditions. The structure is approximately 280 by 127 ft in plan and is located west of the reactor/auxiliary building. The complex consists of two divisions: Division I and Division II. Each division is comprised of a water reservoir, a pump house, a two-cell mechanical draft cooling tower, and two emergency diesel generators. Division I is in the south side, and Division II is in the north side of the complex. Each division has the capacity to safely and orderly shut down the reactor during normal and/or accident conditions completely independent of the other.

The cooling tower system is designed such that it can function even if one tower division is damaged and rendered out of operation (UFSAR 3.5.1.3.2.2).

The RHRSW system transfers heat from the RHR system to the Ultimate Heat Sink (UHS). The UHS includes two 3,465,000-gal reservoirs. Mechanical Draft Cooling Tower (MDCT) fans cool the UHS. The ultimate heat sink design conforms to the requirements of USNRC Regulatory Guide 1.27.

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The Past Operability determination for E1150F068B found that the MOV was unable to perform its design basis functions from May 3, 2017 at 5:48 am, when the RHRSW FCV was last successfully stroked under dynamic conditions, through May 24, 2017 at 4:04 pm, when the RHRSW FCV was returned to service.

The Division 1 RHRSW was available throughout the event except on two occasions. Division 1 of RHRSW was declared inoperable for MDCT Nozzle Cleaning activities on May 9, 2017 from 8:41 am to May 9, 2017 at 11:18 pm. Division 1 of RHRSW was again declared inoperable for MDCT Nozzle Cleaning activities on May 11, 2017 at 8:35 am through May 11, 2017 at 10:01 pm.

The UHS and RHRSW capabilities were analyzed and available equipment would have prevented exceeding design basis limits in the event of a design basis accident.

Each UHS/RHRSW division has the capacity to safely and orderly shut down the reactor during normal and accident conditions completely independent of the other division. The degraded Division 2 RHRSW flow, required an analysis when UHS/RHRSW Division 1 was out of service. Division 1 RHRSW was out of service on May 9, 2017 and May 11, 2017 for maintenance on MDCT Cell sparger nozzles. During maintenance, only a single Division 1 MDCT cell was made unavailable with the other Division 1 MDCT cell left intact and available. The analysis showed that sufficient cooling capability was available for the suppression pool with degraded Division 2 RHRSW flow. The availability of remaining Division 1 equipment coupled with the flow from the Division 2 RHRSW FCV (E1150F068B) would have been sufficient to remove the heat from the Division 2 RHR system.

CAUSE OF THE EVENT

The Direct Cause was Abnormal Stress- Failure of the Anti-Rotation Key (Dropped out of keyway).

The Apparent Cause is System/Equipment Operation – High Vibration. E1150F068B experiences high vibrations during throttling activities to support RHRSW initiation and shutdown. The anti-rotation key tack welds failed and the key dropped out of the keyway allowing the stem to rotate under dynamic conditions.

The first Contributing Cause is inadequate performance monitoring. The MOV Preventive Maintenance procedures do not inspect all anti-rotation devices.

The second Contributing Cause is troubleshooting. The MMA26 "Troubleshooting" pursued Routine Troubleshooting on May 5 through May 7, 2017. The Routine Troubleshooting did not identify a true fault failure mechanism. The limit switches were replaced. The FCV PMT produced satisfactory results which were accepted as completion of troubleshooting and repair activities. The valve was stroked in a static, no RHRSW flow environment without the anti-rotation key. The anti-rotation key failure was not considered as a failure mechanism on May 5, 2017. The troubleshooting performed was inadequate. The potential failure mechanisms were not all considered.

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CORRECTIVE ACTIONS

The anti-rotation key was replaced and the RHRSW FCV passed PMT. Division 2 RHRSW was returned to service.

The anti-rotation key design is used on 17 other Generic Letter (GL96-05 "Periodic Verification of Design-Basis Capability of Safety-Related Motor-Operated Valves") program MOVs at Fermi 2. Walkdowns including keyway weld inspections have been completed. The inspections verified the integrity of the anti-rotation device. There are no MOVs exhibiting the symptoms observed on the RHRSW FCV (E1150F068B) prior to the failure of the anti-rotation key.

Internal and external Operating Experience (OE) was searched. There was no other similar internal or external OE identified that identified anti rotation key welds as a failure mechanism.

The RHRSW FCVs, Division 2 E1150F068B and Division 1 E1150F068A, experience high vibrations during throttling activities to support RHRSW system startup and shutdown. Using System Operating Procedure (SOP) 23.208, "RHR Complex Service Water Systems," which directs the operation of RHRSW FCV E1150F068B during startup of Division 2 RHRSW (Procedure Section 5.4), the E1150F068B is throttled open for 5 seconds for one of the Division 2 RHRSW Pumps. One pump was then started. The RHRSW FCV is then throttled open to obtain 5400 gpm to 6300 gpm as indicated on E11-R602B, "Division 2 RHR HX SW Flow Indicator." After flow is verified, the second pump is started, if available, before throttling E1150F068A (B) fully open. Procedure SOP 23.208 is being evaluated for revision to prevent vibration damage on RHRSW FCV valves E1150F068A and E1150F068B.

Procedures 35.306.003 "Motor Operator Periodic Inspection" and 35.306.020 "Motor Operated Valve Mini Periodic Inspection," will be revised to include inspection steps for anti-rotation devices on Globe Valves.

The functional testing and PMT were ineffective in discovering the failure and its associated pre-cursors on May 7, 2017. The troubleshooting performed was inadequate. The potential failure mechanisms were not all considered. An evaluation will be performed to assess additional training on the expectations for troubleshooting performance.

PREVIOUS OCCURENCES

None.

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ADDITIONAL INFORMATION

Failed Component: Powell, William Company Globe Valve due to high system vibration transferred to the RHRSW FCV.